

Crescent Chique Designs 2.0

SUMMARY

Three-Tier Architecture Frontend Backend API MySQL Database

TECHNICAL SKILLS

HTML5

CSS3

Bootstrap 5

JavaScript

React.js

Python 3.12

Flask

Flask-Login

Flask-WTF

SQLAlchemy

bcrypt

ReportLab

MySQL 8

pytest

GitHub

Redis Caching

DATABASE DESIGN

users: user_id, name, email, password_hash, role, created_at customers: customer_id, user_id, phone, address, city, state designs: design_id, title, description, room_type, style, price_per_sqft, image_url appointments: appointment_id, customer_id, appointment_date, appointment_time, status, requirements quotations: quotation_id, customer_id, design_id, area, material_cost, labour_cost, tax, total_amount projects: project_id, customer_id, quotation_id, project_status, start_date, expected_completion employees: employee_id, name, designation, phone, salary, availability_status project_assignments: assignment_id, project_id, employee_id reviews: review_id, customer_id, project_id, rating, comment

API DESIGN

Authentication: POST /register, POST /login, POST /logout Designs: GET /designs, GET /designs/{id} Appointments: POST /appointments, GET /appointments Quotations: POST /quotation/generate, GET /quotation/{id} Projects: GET /projects, PUT /projects/status Reviews: POST /reviews

SECURITY

Password hashing using bcrypt CSRF Protection SQL Injection Prevention Input Validation Secure Sessions

FILE UPLOADS

Allowed: PDF, PNG, JPG, JPEG Maximum Size: 10 MB Use Cases: Floor Plans, Site Images, Requirement Documents

QUOTATION ENGINE LOGIC

Formula: Total Cost = (Material Cost) + (Labour Cost) + (Design Charges) + (Taxes) Example: 1000 sq.ft × ₹1500
Base Cost = ₹15,00,000 GST = 18% Final Amount Generated

PDF GENERATION

Generate: Company Header, Customer Details, Cost Breakdown, Terms & Conditions, Signature Section Library:
ReportLab

LOGGING

Store: Login Activity, Quotation Generation, Appointment Creation, Status Changes

DEPLOYMENT

Phase 1 Backend: Render Database: MySQL Frontend: Render Static Hosting

TESTING

Unit Testing: pytest Coverage Target: 80%

PERFORMANCE GOALS

Page Load: < 2 sec API Response: < 500 ms Concurrent Users: 100+

CI/CD

GitHub Workflow: Push → Automatic Testing → Deployment

FUTURE UPGRADE PATH

Version 2: React Frontend, REST API Expansion, Redis Caching, AI Recommendations Version 3: Mobile App, Payment Gateway, AR Room Visualization Version 4: Multi-Tenant SaaS Platform, Subscription Plans, Vendor Marketplace